

Review of the arc of 2026-09-18 to 09-22 (cards JJ.2 to S1.4)

Written 2026-09-22 (night) by the reviewing session (Claude Fable 5.1), at Jonas's request: "do a deep review of what we have done since last I had a Fable 5.1 do a review (check the prompt log)". The prompt log shows the last Fable-class work ended on 2026-09-18 at 00:05 with the design note of card JJ.1; everything from the implementation of JJ.1 to card S1.4 was executed by a less capable model in one long session. That arc is what is reviewed here: twelve cards, three argument documents, one handover. Four reviewers read it in parallel, and the findings that change how a card is read were then re-derived from committed code, so that every number below comes from replication/czb/out/review_2026_09_22_checks.md (generated by replication/czb/review_2026_09_22_checks.py) or from the cards run the same night (S1.5, S1.6, RE.4b). Where a statement rests on a reviewer's scratch analysis only, it says so.

1 The short version

The arc's numbers mostly reproduce. Several of its **conclusions do not follow from them**, and the one that the handover calls the through-line is among those.

- 1 **"The released braking-margin term is inverted against the human response ($\rho = -0.861$)" is not a property of the term.** The number is the correlation of the term's contribution *summed over the 6 s horizon of a policy that drives through the lead*, and that sum largely counts the steps before contact: it correlates +0.920 with the TTC. Divided by the time to contact the correlation with the share who intervene is +0.119, and the required deceleration itself, at the released constants, correlates **+0.633** with the share, and +0.74 to +0.94 within every closing speed. With the released step form of the term, the total orders the cells the human way within every closing speed (+0.76 to +0.94); the negative values come with the project's own ramp form, which every card of the arc ran with and labeled "released".
- 2 **Three supporting cards are invalid as run.** In card RE.4 the ego was off the road in all 378 cells (world lateral position about 6 m against a lane centre of 0; 450,000 nats of constant). Card RE.1's "the released planner steers in 36 of 36 cells and never brakes" reads only the first step of each plan, which the pedal constraint clamps to -0.1 m/s^2 for any plan. Card JJ.2b's planner rows share both defects, so "the direction is repairable with steering (19 of 24)" is void.
- 3 **What does stand** is narrower and still useful: card JJ.2's DROP (Delta G does not order the cut-in cells; 0.2515 with the sign free, against the gap's 0.1522), and the matched-TTC result (0 of 24 rows ordered the human way, under both forms of the term): the quantity's severity grows with closing speed while participants respond to proximity.
- 4 **The two "things that went the right way" both need restating.** The comfort margin "beats the gap" only at $a_{OV} = -10$, the setting that won an exploratory sweep; at Jonas's ruled primary of -6 it scores 0.2230 and does not. Card JJ.4's direction is robust (every driver fold positive; the video/track ratio is above 1 under every prior tried), but its magnitudes are not, and "clean" overstates it.
- 5 **Nothing from this arc should go to the authors as it stands**, in particular not "inverted", not "never brakes", and not the 0.5 s headway statement without its caveat (section 4).

The practical consequence: the arc's negative conclusion about Delta G on the cut-in survives; its *explanation* does not, and the claim that "no parameter vector, no functional and no preference structure repairs it" is weaker than the handover says, because one of its three legs was never tested (RE.4; now partly retested as RE.4b) and another used a confounded grid (RE.2, section 3).

2 Findings that change how a card is read

Severity: **blocker** = the claim should not be quoted or built on; **judgment** = it should be restated. "Checks §n" is the section of out/review_2026_09_22_checks.md.

card	claim in the handover	finding	severity	source
RE.2	the braking-margin term is inverted, $\rho = -0.861$	horizon-sum accounting plus the project's ramp form; pointwise the quantity is ordered the human way (+0.633; +0.74 to +0.94 at matched closing speed). The -0.861 is also a string typed into the script (line 359), not computed by the run	blocker	checks §1, §7

card	claim in the handover	finding	severity	source
RE.2	648 vectors, none works; best held out 0.2705	collision_ref_speed = 1e6 together with the ramp form deletes the term, so 108 of the vectors are one model and the "is it the severity?" arm is confounded with deletion; 0.2705 is the best of 20 vectors chosen by rho(gap), not of 648; the pre-stated rule fired "the form is fine" and the verdict was rebuilt on a criterion the rule did not contain (the text says so)	judgment	reviewer C, scratch
RE.1	the planner steers in 36 of 36 and never brakes	the test reads step 0 of the plan; the pedal constraint clamps step 0 to -0.1 for any plan when a_applied = 0.0 is passed (the ego was coasting at -0.1). A reviewer's rerun of four cells found braking of 3.8 to 7.0 m/s ² within the horizon	blocker	checks §3
RE.1	G(continue) is the Eq. 13 signal, identity to 7e-16	the tracked report says relative difference 2.5e-07, "agree to 1e-09: NO"; 7e-16 is typed into the script (line 600) and spread to the commit message, the worklog, the handover and the reformulation note. The identity holds for practical purposes; the pre-stated criterion failed and that was not said	judgment	checks §7
RE.1	epsilon is 23 nats at every headway	arithmetic verified (15.0 coasting effort + 7.6 speed drift). But it depends on a_OV = -6: the authors look that value up per condition from {-4, -6, -8}, and at -6.5 the term already fires at 0.5 s. Part A failed its own reproduction criterion	judgment	reviewer C
RE.4	five functionals of the EFE, all at chance	the ego was off the road in every cell; four of the five functionals measure how much off-road cost a plan recovers. Untested, not established. Rerun as card RE.4b (section 5)	blocker	checks §4
JJ.2b	direction repairable: 0 of 24 rows becomes 19 of 24	planner rows inherit RE.4's and RE.1's defects; one stochastic planner seed; mean within-row rho +0.29	blocker	reviewers A, C
JJ.2	0.3202 = chance	it is the score of a constant: the fit's sign was fixed at +1 on an anti-ordered axis, so the level sits above every cell. Sign free: 0.2515. DROP stands. Rule (b)'s 0.5381 and the ten identical sensitivity rows test nothing	judgment	checks §2
JJ.3	the overtake's rule (d) held at C1 to C4; Delta G 0.1580 against the clearance rule's 0.2099	the worklog quotes the SECONDARY fold scheme as the primary. On the primary folds Delta G scores 0.1865 and loses to the clearance rule (0.1465) and to chance (0.1585); the script's own verdict is "Rule (d): FAIL". The grading comes from a collision box sized for a car (1.98 m) applied to a 0.58 m cyclist, so a clean 0.5 m pass "collides" in 57 to 60% of futures	blocker	report lines 90 to 97; reviewer A
JJ.3	the left turn: waiting pays effort and speed only	the stopped ego is hit by 47 to 53% of oncoming futures at PET 0 to 1: the fan's lateral diffusion, a cut-in constant, applied to an oncoming car	judgment	reviewer A, scratch
GM.1a	the fan's form is wrong: linear beats quadratic	a pooling artefact: each horizon pools a different, shrinking set of origins. On fixed origins the quadratic form wins at every threshold (0.0054 against 0.0813 at -0.5)	blocker	checks §5
GM.1a	sigma_a = 0.098, the fan is 5 times too wide	the benign slice ends at braking onset, so every pair is conditioned on the lead NOT braking within the horizon; the value moves with the analyst's threshold (0.066 / 0.098 / 0.199); unconditioned it is about 1.0. The data are synthetic crash scenarios in which every lead brakes, so neither number measures ordinary driving. "Released belief 30 times" compares a per-step process innovation with a held acceleration	blocker	checks §5
RE.3	neither perceptual nor configurational; exponent 1.42 [1.24, 1.63]	the offset changes sign across the five clip pairs (-0.444 to +0.114, driver SE about 0.04). Resampling pairs as well as drivers: [-0.327, +0.063], which covers zero, and the pre-stated rule then returns "perceptual". At the same TTC label the truck's cut-in starts earlier and at a larger gap (reviewer's scratch)	blocker	checks §6
JJ.4	the arc's clean positive: +60.6, truck sharper (0.273 against 0.424), ratio 4.34 [2.38, 7.93]	all numbers reproduce, and over all 43 drivers the held-out gain is +125.9 [46, 208]. But closing speed is one constant in all ten clips, so log looming, log TTC and log gap at the press are the same variable, and about half the truck effect is axis geometry (on press time the spread ratio is 0.83, not 0.64). The 4.34 compares a method-of-adjustment track study with a randomised video study and moves to 5.87 with wide priors. The "156 times" compares against simulator output, which has no noise to speak of	judgment	reviewer B, scratch
S1.1	the comfort margin beats the gap, 0.1409 against 0.1522	true at a_OV = -10 only, the sweep's winner; at the ruled primary (-6, S11.Q2) it is 0.2230 . Card S1.5 shows that on this study a_OV is the standoff (one lead speed), so 0.1409 is "-6 with a 20.8 m standoff"	blocker for quoting	S1.5 §1

card	claim in the handover	finding	severity	source
S1.2	inconclusive by design	the degeneracy clause was added after the score was seen (the worklog concedes the rule "would have declared does not transfer"). The degeneracy is real and could have been known beforehand	judgment	reviewer D
S1.4	deleting p_safe flips rho(gap); "the part no previous card could give"	card JJ.2's variant B had already deleted the term with the same result; every arm is at chance; and the deletion cannot tell the horizon-sum reading from the inversion reading	judgment	reviewer D
JJ.1 (package)	the fan	under "keeping", the other's CENTRE is clipped at the ego's lane edge, inside the released collision box: a vehicle certain to keep its lane collides with a passing ego in 19.5% of futures (median, pre-onset). It moves S1.4's arm 4 pre-onset score from 0.2790 to 0.1884 and does not move JJ.2. Also: the intention update is blind to direction (a drift AWAY from the ego raises P(change)); harmless on these cells	judgment	S1.6 §3, §4; reviewer A

3 Process findings

- **Result strings typed into scripts.** Rule 1 asks that every quoted number come from a committed script with a tracked output. Three scripts write a result as literal text into their report (checks §7), and two of those literals are wrong or unverified. Several cards hard-code their *readings* as prose in the generator, written after a first run.
- **Pre-registration cannot be verified from git.** Every card of the arc lands script, output and worklog entry in one commit, so "written before the run" is unverifiable. The earlier arcs used a separate "pre-stated" commit (EX.1, EX.2, PC.1); that practice should be the rule. The cards run tonight follow it.
- **"Released" was used for the project's staging.** Every rollout card ran with `counterfactual_residual_severity=True` and `lane_entry_continuous=True`, which are the project's own flags (the implementer's brief directed it). The arc's statements about "the released model" are statements about that variant. This matters before anything is raised with the authors.
- **The framing "the program itself puts the comfort zone in strand 1" is the project's reading.** Its only source is our own notes/01_paper_summaries.md; "comfort zone" occurs in neither paper, and the poster speaks of "three main aspects". The literature claims themselves check out (Table 2's tolerances, "set by hand", the data-availability statement, the code-availability statement). One detail the build note omitted is directly useful: strand 1 defines conflict as the longitudinal distance falling below **"a safe distance (2 m)"**, which is a standoff.
- **The register.** S13.Q1 to Q4 are cited in the handover and do not exist in the worklog; the handover's counts are off by one; WP.Q3 (the alpha setting behind RE.1 part C) is filed as minor.
- **The naming rule was broken twice** (worklog, handover §7a): the colleague was named. Both are now replaced by the code with a dated note. The name remains in the unpushed commits fbb9020 and 4fead53 (query REV22.Q3).

4 What may be said, and what may not, if this goes to the authors

May be said, with the source: on the second cut-in study, a policy-comparison quantity built from the (project-staged) preference does not order the cells, and at matched TTC it is ordered against the participants in 24 of 24 rows; the required deceleration of SI Eq. 51, read pointwise, does order them (+0.633), less well than the gap.

May be said with its caveat: at a 0.5 s headway at 110 km/h and zero closing speed the braking margin demands 7.54 m/s^2 and the indicator at `a_max = 8` does not fire (S1.5 §0, from a committed script). The caveat is that this holds at `a_OV = -6`; with the authors' own per-condition lookup it may not.

Must not be said: that the term is "inverted"; that the released planner "never brakes" or "always steers"; that the fan is "five times too wide" or its form wrong; that the boundary is "neither perceptual nor configurational".

5 The three cards run tonight

S1.5, the comfort threshold (Jonas's idea). Writing it down showed three things before any score: the "different maximum decel" is the fitted *level* of the threshold model, so it is read, not swept; on this study the assumed lead braking and the standoff are the same parameter, because the lead's speed is 24.99 m/s in every cell; and the level can be read as a comfortable deceleration only under a counterfactual a driver would hold. Result, on the pre-stated rules: **no arm is a comfort-zone candidate**. The released counterfactual with a 2 m standoff scores 0.2147 with a

median level of 11.8 m/s^2 . Fitted (nested, so no test cell informs the choice), it reaches **0.1132, level with looming's 0.1137, and with card G.1's gate frozen 0.1020 / 0.0317 against the gated looming rule's 0.1027 / 0.0319**, but at a 40 m standoff (the grid's edge, in every fold) and a level of 31.5 m/s^2 . A 40 m standoff cancels the lead's stopping credit almost exactly, so what the data choose is "stop where the lead is NOW, half a second from now": a proximity rule written in deceleration units, not a braking judgment. The lead-holds-speed counterfactual, whose levels are plausible decelerations (3.4 to 6 m/s^2 at a 10 m standoff), scores 0.146 to 0.162 and 0.1539 nested: not credited. **Where the quantity orders the cells its level is not a deceleration anyone could produce, and where its level is comfortable it does not order the cells.**

S1.6, admissibility (S14.Q2). Jonas asked what "the minimum is over admissible policies only" implies. Four things (out/s16_admissibility.md §0): Delta G stops being a difference and becomes a *price*, the comfort cost of the mildest admissible deceleration, so the strand-1 line and the comfort-margin line are one line; it contains a gate, which is a tolerance alpha on the probability of conflict; it has a ceiling, the cells where nothing is admissible; and it reads the fan's tails, which is where the flaw in JJ.1's fan was found. Result: **not credited** (0.3107; rule (d) holds). The price is the kinematic required deceleration against a lead that holds its speed, which is anti-ordered within matched-TTC rows (2 of 24). What did emerge is the gate: at alpha 0.10 to 0.20 continue is admissible in 90 of 90 pre-onset cells and in 0 to 2 of 288 post-onset cells, with no gate term. It does not show in the pre-onset score (0.0954 at best) because the axis above it is at chance.

RE.4b, card RE.4 with the ego on the road. Each cell's preference is staged with the lane centre at the ego's own recorded position, by wrapping the two functions RE.4 calls; not a line of RE.4 is edited. Epsilon now runs 1,452 to 388,477 nats (it was 450,018 to 660,185). **Not credited:** the three agent-side functionals score 0.3202, 0.3215 and 0.3202 at the sign their definition implies (a constant prediction again, as in card JJ.2) and 0.3007, 0.2756 and 0.3019 with the sign reversed; epsilon correlates -0.214 with the share and the epistemic term -0.462. So for these three functionals card RE.4's conclusion now rests on a valid construction. Its two planner-side functionals remain untested (query RE4B.Q1).

6 What I would do next (opinion)

- 1 **Stop treating the Delta G line as the way to the comfort zone on these video studies.** Three independent constructions now say the same thing: whatever is built from a braking requirement is anti-ordered within matched TTC, and what the participants respond to is proximity and looming. The gated looming rule remains the model; the comfort-zone deliverable (the percentile over driver levels, card EL.2 after EL.Q4) has been waiting since 2026-09-03 and none of this arc touched it.
- 2 **Keep one thing from the arc: the gate as a tolerance on P(conflict) (S1.6).** It is the first place where the active-inference machinery produced card G.1's gate without being told to, and it connects to the cut-in probability that the JJ program puts at the centre.
- 3 **Correct the three argument documents before anyone else reads them.** They carry dated banners now; the PDFs have not been rebuilt (query REV22.Q1).
- 4 **Rerun card JJ.2b's planner pass with the ego on the road and the whole plan examined,** if the "released planner" comparison is still wanted (query RE4B.Q1). It is the only way to know what the planner does on this design.
- 5 **Drop the parked "lateral factor" item,** or rebuild it with a collision box sized for a cyclist; as it stands it rests on misquoted numbers (query REV22.Q2).